

User study: can fencers read their own ratings?

Status: protocol drafted, no sessions run yet.

This file is split into two stages on purpose. The protocol below goes into the repository before any sessions run, so the git history shows the plan came before the data: the research questions, tasks, and measures were fixed in advance, not affected by the findings. The findings will be added in Part 2 after the sessions, following the same protocol. The document is also frozen as an OSF preregistration before the first session: git history and a registration is easy to follow. Any post-pilot changes appear here as revisions and are noted against the registration, which itself does not change.

Part 1: Protocol

1. Why this study

The design choices in [DESIGN.md](#) leave me with a set of questions I can't answer from usage data or my own judgement, because they're about what's going on in a fencer's head. The first three rounds of beta feedback (on [FEEDBACK.md](#)) were reactive. People told me what looked wrong, which helped me find bugs but not grasp the problems. Someone who misreads their rating will never file an issue about it.

The study focuses on the open questions the documentation already notes:

- **RQ1 (low end ratings).** Do fencers understand that the shown number is at the low end of the uncertainty, and that it increases with more data, instead of just wins? I noted in [DESIGN.md](#) that this is “a design decision I’m still not certain about.” Round 3 feedback (“the elo is too close together”) may partly be a sign that people are reading it wrong.
- **RQ2 (pool/DE split).** Can fencers decipher the two ratings and connect them to a training decision?
- **RQ3 (age-category defaults).** Does the default “All ages” leaderboard mislead people about where they stand in respect to selections? A Round 3 report said exactly this.
- **RQ4 (tableau + DE line difficulty).** Are fencers able to read the rebuilt tableau, and do they understand what's meant when the site says “line average” and “sweep odds”?
- **RQ5 (calibrated trust).** Do fencers trust the ratings properly? Neither ignoring them nor reading too much into them? Does seeing the Method tab affect that?

- **RQ6 (club section tab).** Round 2 feedback suggested that “where should I fence?” may be more useful than “how strong is this club?”. Does the current page answer the question fencers actually have when using it?
- **RQ7 (profile login + data-flagging).** Can the user log into their own profile, dispute incorrect data, and understand what happens next?

2. Method

Column 1	Column 2
Design	Moderated remote usability study, with a concurrent think-aloud
Participants	6 to 8 NZ fencers (see allocations below), plus 1 pilot
Format	Video call with screen share; participant drives on their own device
Length	60 minutes per session
Recording	Screen and audio, with consent. Structured notes as a fallback if declined
Moderator	Me. I am also the developer. Section 11 addresses the bias this creates
Analysis	Task success and comprehension scoring per participant. Thematic grouping of observations. Severity-rated issue list, as per-RQ synthesis

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I will run the sessions on the current site, because the study's questions concern the participant's own data. One task will use another fencer's public profile instead, which I mention later.

3. Participants

Sampling quotas

Eight slots, looking for spread over representativeness. With a sample this small, the goal is to hear from every kind of reader the site has, not to estimate population parameters.

Column 1	Column 2
Dimension	Target spread
Competitive level	At least 2 nationally ranked in the top 10, at least 2 in the middle, at least 1 relative newcomer (less than 2 years fenced)
Age category	At least 1 junior (18 or 19), at least 2 open/senior, at least 1 veteran
Weapon	All three weapons represented across the pool
Gender	At least 3 of each
Region	At least 2 outside the main centres (Christchurch, Wellington, Auckland)

Relationship to the system	At least 1 fencer whose rating here is below their official FeNZ ranking, recruited knowingly.
Selector view	At least 1 participant who has been on a selection panel

The rows are not equally weighted, and deciding that now is better than changing scope mid-recruitment. Two are required: the competitive level spread, and having a sceptic, because they do work for RQ1 and RQ5. The rest are soft targets, looked for during recruitment, but relaxed if scheduling requires. Will be reported in Part 2.

Recruitment

I'll post in the 'NZ Fencing Community' Facebook, as well as DM people to fill gaps. My post will give the details of the study: improving the site, length, remotely held, recorded (with consent), who I'm looking for, and link to an eligibility screening, which asks age (must be 18+), weapon, years fencing, region, if they've used the site before, and whether they are part of the selection panel. I'll pay the participants \$30.

The sceptic is directly invited. The message will say that the site currently ranks them below their official ranking and that I want to hear from the people that it is most harsh on.

Inclusion/exclusion criteria

- **18 or over.** Cadet fencers are a large user group, and this is a big limitation (as discussed in section 11), but running minors through a recorded study requires guardian consent as well as safeguarding procedures that go past what I can do myself. The junior perspective is mostly represented by 18 to 20-year-olds, who fence in junior events.
- Must appear in the dataset with at least one competition, except that the newcomer slot may be filled by a fencer with minimal data, since "the site has nearly nothing about me" is itself a state worth testing.
- No immediate family and no one who contributed to the project.

4. Ethics + consent

This is an independent project without a review board, which means a higher standard for self-discipline rather than lowering it. The procedures below follow standard human-subjects practice for minimal-risk usability research.

Consent will be collected twice. Once in writing before the session (a typed "I agree" reply in response to a consent summary is kept), and once verbally on the recording at the start of the session, using the script in section 8.

What participants are told:

- What the study is for and who is running it (me, personally; not FeNZ).
- The session is recorded (screen and audio) if they agree. Declining recording does not exclude them, it just means I take notes instead.
- They can stop at any time, skip anything, and withdraw afterwards up to the point findings are published, in which case their recording and notes will be deleted. "Published" means one thing: the commit of Part 2 to this public repository. That commit is the withdrawal cutoff.
- The site is being tested, not them. There are no wrong answers, and struggling with a task is the most useful thing that can happen.
- Anonymised findings may also help future academic work, including a possible university thesis. Consent covers that use from the start, so nobody is asked later to give permission they thought they had already given.
- What happens to the data (below).

Data management. All of the recordings will be kept in a private locally stored folder plus one backup, they won't be committed to the GitHub repo, and will be deleted within 6 months of the findings being published. Notes and quotes are going to be anonymised in the write-up: participants will be coded as P1 to P8 with rough descriptors (weapon, level, category). A participant's ratings shown on screen during their session. This is already public data, but my findings will avoid connecting a P-number with any identifying details. One task will be stricter: the head-to-head task records participants' estimates of specific clubmates' chances of winning. The write-up will never link a wager answer to a participant, even under the rough descriptors.

There is Emotional Risk. Ratings are measures of someone's skill, and the site might show someone a lower number than they'd expected. The consent summary talks about this, and the debrief has space to talk about it.

5. Session plan

Segment	Minutes	Content
Welcome and verbal consent	5	Script section 8.1; recording starts only after agreement

Background interview	6	Welcome and verbal consent Questions below
Think-aloud practice	2	One neutral warm-up (their club's website or a news page) to practise narrating
Tasks 1 to 6	35	Section 6; fixed order, rationale below
Trust probe and Method tab	5	Task 7
Debrief and SUS	7	Section 8.4 plus the 10-item System Usability Scale

Background interview questions (conversational/general):

1. How long have you been fencing, and what does your competition year look like?
2. When rankings come up, where do you look things up today, and what for?
3. Have you used askFRED, fencingtracker, or similar tools?
4. Have you seen this site before? How often, and for what? (Gives an estimate on 'freshness'.)
5. (Before opening site): What do you think the site's rating means?

Task order will be fixed. The tasks have a naturalistic narrative order (profile > field > system's honesty about itself) that I think produces good think-aloud. The downside (small order effects) is discussed in section 11.

Per-participant prep, before sessions are held: confirm their data is accurate and up to date. Choose the competition (their most recent), and a fallback to a larger event if they haven't competed recently. Choose the selection bracket that applies to them, and check their site rating against their official FeNZ ranking to anticipate any emotional risk.

6. Tasks

Each task outlines the exact scenario, what it's testing, the success criteria, and some follow-up probes. Probes are asked after the participant finishes or gives up, never during, except the standing think-aloud nudge ("what are you thinking?"). Every task closes with the single-ease question, "Overall, how easy or hard was that?" on a 1 to 7 scale, asked before any probing.

T1. Your own numbers (RQ1, RQ2), 8 min

"Find yourself on the site, and walk me through what the site is saying about you as a fencer, as if you were explaining it to a clubmate."

- **Tests:** name search, profile understanding, the conservative rating, the range, pool/DE split.
- **Success:** full = finds profile and correctly explains that the headline is the lower end of the uncertainty range and that pool and DE are separate ratings/skills.
- **Partial** = locates the profile, reads the numbers, but treats the displayed number as "my exact rating" or cannot say what the range is.
- **Fail** = cannot read the profile at all.
- **Probes:** "There are two numbers, and a range next to each. What is the range?" "If you fenced next weekend and won everything, what would happen to these?" "If you didn't fence for a year, what should happen?"

T2. The training read (RQ2), 4 min

"Based on what your profile shows, if you had one extra training session a week, what would you spend it on, and what on the site tells you that?"

- **Tests:** whether the pool/DE split supports the decision it was built for.
- **Success:** full = connects the weaker rating, or the field-overview pattern, to a concrete training focus (e.g. "I would fence more pool bouts, I would fence more DE bouts").
- **Partial** = picks something but cannot ground it in the page.
- **Fail** = the page offers them nothing.
- **Probe:** "Is that something you already knew about your fencing, or did the page tell you something new?"

T3. Where you rank (RQ1, RQ3), 5 min

"Imagine [NZ junior team/worlds team/veteran worlds team] is being picked for your weapon. Find where you're currently ranked against the people you'd be competing against for that spot."

- **Tests:** the All-ages default, the age-category filter, gender and weapon pills, min-bouts comprehension.
- **Success:** full = arrives at the correct category leaderboard and their true position.
- **Partial** = finds a leaderboard but the wrong pool.
- **Fail** = user gets lost and can't find the correct leaderboard.
- **Probes:** "Who was in the first list you found?" "Is this list who you expected? Anyone missing or surprising?" (Surprise answers here feed RQ3 directly.)

T4. Reading a competition (RQ4), 7 min

"Open (the most recent competition the participant fenced). First, how did your day go, according to the site? Second, who had the hardest run to the title, and how would you know?"

- **Tests:** competition results view, field overview, the tableau, line average and sweep odds.
- **Success:** full = finds the tableau or toughest-lines table and explains line average and sweep odds correctly.
- **Partial** = finds the tableau and can read who beat who, but doesn't understand the line difficulty or sweep odds.
- **Fail** = can't navigate the competition view.
- **Probes:** "Which is the hardest line? What makes it the hardest one?" "The tableau says 'sweep odds is n'. What does that mean?" "Is this what you would expect?"
- **Note:** if the participant's bout history is too limited (probably the newcomer slot), use the set fallback competition.

T5. The prediction (RQ5), 5 min

"Pick two fencers you know well and compare them head-to-head. The site will give a prediction. Would you put \$20 on it at those odds? Why? Why not?"

- **Tests:** understanding of head-to-head and user's trust in the site. The wager question invites a proper reading rather than a polite one.
- **Success:** understanding only. Full: reads the per-stream probabilities accurately, including that it's only a prediction from ratings, not a guarantee. Partial/fail accordingly. The trust answer itself is just data, it shouldn't be scored.
- **Probes:** "What would make you trust this number more?" "What does the site know that you don't? What do you know that it doesn't?"

T6. Something is wrong (RQ7), 4 min

"Imagine you spot a bout on your record with the wrong score. Show me what you'd do about it, right up to the point of sending, and stop there."

- **Tests:** ease of discoverability of sign-in and the bout dispute features. (The participant will sign in with their own licence number if willing. If not, they go to the sign-in page and the moderator will screen-share the remaining steps from a test profile, with instruction).
- **Success:** full = participant can find the sign-in page and get to the dispute form on the correct bout without any help.
- **Partial** = needs a hint to get there.
- **Fail** = does not discover the option exists.
- **Probe:** "What did you expect to happen after submitting?" (Tests whether the review-queue model is understood.)
- **Note:** nothing is submitted during sessions. If a participant spots a genuine data error, it is logged and handled through the normal flow after the session.

T7. The system grades itself (RQ5), 5 min

Before opening it: "On a scale of 1 to 7, how much do you trust these ratings to reflect real current ability? Hold that number." Then: "The site has a Method tab that reports how accurate its predictions have been. Take a look and tell me what you make of it. Now, same question: 1 to 7."

- **Tests:** whether the accuracy panel and FAQ are legible to a regular user, and whether evidence has any effect on trust.
- **Success:** full = at least one correct answer about the data, without any help.
Partial: reads and understands the FAQ, but the dashboard isn't fully understood.
Fail = bounces off entirely.
- **Probes:** "What's the most convincing thing on this page? The least?" "Who do you think this page is for?"

T8 (optional, if time). Where should I fence? (RQ6), 4 min

"A friend is moving to Auckland and asks where they should fence. Can this site help you answer them? Does your answer change depending on what weapon they fence?"

- **Tests:** whether the Clubs section answers the question users bring to it.
- **Success:** descriptive only. The expected result, based on Round 2 feedback, is a mismatch between the page's "how strong" framing and the task's "where should I go" question. Observing how participants improvise around it will help the redesign.

7. Measures

- **Success:** full/partial/fail per task, criteria as stated above, scored from recordings after the session.
- **Per-task ease:** the single ease question, 1 to 7, immediately after each task. It costs ten seconds and, unlike SUS, localises difficulty to a task rather than to the site. Like SUS, it is comparable across future rounds.
- **Understanding the score:** three things that will be scored from 0 to 2 each (0 = wrong, 1 = partial, 2 = solid) from the T1/T4/T5 answers: the rating range, line average or sweep odds, and the win prediction. This gives me a 0-6 score of how well the system's ideas held in practice. The developer will also be the scorer. Two participants' sessions will be recoded by someone not part of the project, with agreement conveyed and any disagreement discussed in the write-up.
- **Trust:** the two 1-7 ratings on the Method tab, reported by the participant, without any averaging. The before/after design can pressure people into giving two different answers (as mentioned in section 11), so the result should measure the direction of change and the participant's reasoning, not the size of the change.
- **SUS:** the 10-item scale. With a sample of 6-8, the score is only a rough measure. It is included because it is comparable across future rounds of this same study.

- **Qualitative data:** Includes notes by session (with timestamps), including quotes, user's uncertainty, mistakes made, workarounds used, and any surprise or doubt.

Analysis: I won't use any inferential statistics because the sample size is so small, and I won't claim anything like "users prefer X to Y". Findings are documented as "P2, P5 and P6 did/said X". Usability problems will be given a severity rating (0-4), based on impact, and how many people had the same issue. Anything rated 3+ blocks or reshapes a v1.1 item. Disconfirming evidence is reported with the same prominence as confirming evidence, per RQ, including the possibility that a design I like (the conservative display) simply does not communicate.

8. Moderator script

8.1 Opening + consent (to be read word for word)

"Thanks for doing this. Quick recap of what you agreed to: this is a 60-minute session to test the fencing stats site. I'm going to be recording screen and audio, and the recording will stay private with me and be deleted after I write up anonymised findings. You can skip anything, stop at any time, or tell me afterwards to delete all of it. Two things to remember. First, I'm testing the site, not you. If something's confusing or broken, it's the site's fault, not yours. Finding issues is the whole reason I'm doing this. Secondly, I built the site, but you can't hurt my feelings. Critical feedback is really important, and being polite to me would actually waste both of our time. Please think out loud as you go, say what you're looking at, what you're expecting, and what surprises you. Good to go?"

8.2 Rules throughout tasks

- The only mid-task prompt is a prod: "What are you thinking?" or "What did you expect there?"
- Don't explain a feature, name a page, or point at a part of the screen during a task.
- If a participant is stuck for 90 seconds or asks to move on, mark the task, give a small hint, and continue, or skip the task.
- If the participant criticises something, say "say more about that", never any sort of defence.
- If data looks wrong during the interview, don't stop to debug.

8.3 Probe phrasebook (neutral forms)

"Tell me what you just did." / "What does the number mean?" / "Where would you look next?" / "Is that what you expected? What did you expect?" / "Say that back to me like I'm a clubmate" / "What would you change about this, if anything?"

8.4 Debrief

1. "In one sentence, what does this site's rating mean? (Same question as the intro. The pair of answers is a before/after on the session itself.)

2. "What would make you come back to this site?"
3. "What's the one thing you'd fix first?"
4. "Was there anything from your own numbers today that didn't feel right, or that you'd like to talk about?" (Emotional check from section 4.)
5. SUS questionnaire (link sent in chat).
6. "Was there anything I should've asked and didn't?"

9. Pilot

I will run a full pilot session/practice run with someone who can't be a part of the main sample (probably a friend). The pilot will ensure the timing and logistics work properly, and that any task's wording does not lead participants. Pilot data is used to fix the protocol, not generate any findings. Protocol changes after the pilot are committed as a modification to this file, so the record of what changed and why is public.

10. Reporting

Findings go in Part 2 of this file: a session log summary (participant table with anonymised descriptors), RQ findings with quotes, the severity-ranked issue list, the comprehension and SUS numbers, and a "what will change in v1.1" section where every design choice links to the data that informed it. Anything the study finds that goes against a documented design decision will also be noted in the relevant section of [DESIGN.md](#).

11. Limitations

These are written now, so the write-up cannot skip them:

- **The moderator is also the site developer.** Social desirability pressure exists with the disclaimer in 8.1. So, we should expect people to give polite answers.
- **The T7 trust/confidence probe is insufficient because it pressures people to change their ratings.** Asking people to rate, read the evidence, then rate again is asking for two different numbers.
- **n = 6 to 8** finds understanding failures and usability issues. It won't rank preference or estimate frequency.
- **18+**, so cadets, are not represented.
- **Recruited from an existing community**, most people in FeNZ are aware of the project. The only 'fresh' user is the newcomer.
- **Fixed-task order** swaps order effects for narrative consistency.
- **One 60-minute session** only measures the site's usability at first use, not its value across a longer time period. The first round is deliberately constructive and independent. A second, evaluative round against the v1.1 redesign is planned for 2027 under institutional oversight + ethics review, and a diary-style longitudinal component belongs there, not here.

Part 2: Findings

Not yet run.